

Using statistical learning models with household survey microdata

Applications to energy demand, energy poverty, and environmental taxation

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EAERE

European Association
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Recent applications of AI in Environmental and
Resource Economics



Session 3

Applications and policy implications

Literature example 1: ML for energy poverty with HBS

Italy and household-budget survey prediction

- Scandurra et al. use official Household Budget Survey data for Italy to classify energy poverty with ML algorithms.
- The setup is close to the course workflow: HBS microdata, binary classification, flexible algorithms and class-imbalance concerns.
- Useful extension idea: compare logit, Random Forest and XGBoost, then ask what policy action follows from better classification.

Scandurra, G., Carfora, A., Thomas, A., & Camporeale, C. (2026). Predicting energy poverty using household budget survey: A machine learning approach. *Annals of Operations Research*.

<https://doi.org/10.1007/s10479-026-07187-w>

Literature example 2: household electricity consumption

Survey-based ML for energy-demand classification

- Moon uses a Korean household energy survey with SVM, Random Forest and decision-tree classifiers.
- The study combines household, housing and appliance variables, similar to our feature logic.
- Course connection: our example treats electricity demand as continuous regression; students can discuss when classes are useful.

Moon, J. (2023). Understanding the characteristics of residential electricity consumption in Korea: Applying machine learning techniques to household-level energy survey data. *Journal of Climate Change Research*, 14(5), 569–586.

<https://doi.org/10.15531/KSCCR.2023.14.5.569>

Literature example 3: small-area estimation with random forests

From survey samples to granular poverty maps

- Frink and Schmid propose generalized mixed-effects random forests for small-area estimation of poverty rates in Mexico.
- The paper is relevant because it combines survey-based poverty indicators, flexible ML and geographic hierarchy.

Frink, N., & Schmid, T. (2024). Small area estimation with generalized random forests: Estimating poverty rates in Mexico. arXiv.

<https://doi.org/10.48550/arXiv.2406.03861>

Atlas of Energy Poverty in Spain

A real policy-relevant example using household surveys, modelling and geospatial indicators

Atlas of Energy Poverty in Spain

Household surveys + statistical learning + geospatial indicators

<https://api.otea.info/storage/2026/05/22/c3900990b259cba4e5cd5c57d7083b5a74196d92.pdf>

Thank you

Questions and discussion

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